TCP SOCKET APPLICATION-3

Objective:

Here the objective is to communicate between **TWO** processes over a network i.e.., **SERVER** and **CLIENT ,** with help of **SOCKET** programming by using **TCP / UDP** protocols.

TASK:

Once connection established between SERVER and CLIENT , first server should send one menu MSG to client and it should contain like below.

1. prime 2. palindrome 3. factorial of number

Then , client will give reply back to server depends on requirement.
Then later server should ask for input and client will give reply to server.

Finaly , server will give result to client.

If any wrong info given by client then server should give error MSG to client.

For example1:

Server: 1. prime 2. palindrome 3. factorial of number
Client: prime.

Server : How many ?
Client : 5

Server: Starting value ?
Client: 20

Server: 23, 29, 31, 37, 43.

(should close connection b/w server and client)

For example2:

Server: 1. prime 2. palindrome 3. factorial of number
Client: palindrome.

Server : How many ?
Client : -5
Server: **invalid requirement.**

(should close connection b/w server and client)

For example3:

Server: 1. prime 2. palindrome 3. factorial of number
Client: palindrome.

Server : How many ?
Client : 6

Server: Starting point ?
Client: 100

Server: 101, 111, 121, 131, 141, 151

(should close connection b/w server and client)

For example4:

Server: 1. prime 2. palindrome 3. factorial of number
Client: factorial of number

Server : number please
Client : 5

Server: 120

(should close connection b/w server and client)

Requirements:

Your **TASK** requires **TWO** Processes. Here Process1 is **SERVER** and Proccess2 is **CLIENT.**

Use separate thread for each task.
Ex:
For prime operations use , **prime_thread().**
For palindrome operation use , **palindrome_thread().**
For factorial operation , **fact_thread().**

SERVER always should behave as demon process (It should not print anything on screen except CLIENT INFO).